

EXECUTIVE SUMMARY: RESERVOIR STAGE FITTING & VOLUMETRIC EXPOSURE (LAB 02)

Title: Executive Summary: Reservoir Stage Fitting & Volumetric Exposure (Lab 02)

Author: Civil Engineering Research Team

Date: August 25, 2026

1. MODEL FORMULATION

$$h(t) = c + \frac{a}{1 + e^{-k(t - t_0)}}$$

Physical Constants & Parameters:

c : Lower Datum / Baseline Stage

a : Total Stage Rise (Asymptote)

k : Logistic Growth Rate

t_0 : Inflection Time (Peak Inflow)

2. PARAMETER SIGNIFICANCE & ESTIMATION

Parameter	Description	Estimate (b_i)	Standard Error (SE)	t-Statistic	p-Value
c	Baseline Stage Datum	14.4512 m	0.0124	1165.42	< 0.0001
a	Asymptotic Stage Height	5.7289 m	0.0215	266.46	< 0.0001
k	Kinematic Growth Coefficient	0.6704 hr ⁻¹	0.0089	75.33	< 0.0001
t_0	Inflection Time Threshold	29.2500 hrs	0.0182	1607.14	< 0.0001

3. GOODNESS-OF-FIT SUMMARY

COEFF. OF DETERMINATION

$$R^2 = 0.9978$$

STD. ERROR OF ESTIMATE

$$s = 0.0956 \text{ m}$$

SUM OF SQUARED ERRORS

$$SSE = 0.8412 \text{ m}^2$$

TOTAL SUM OF SQUARES

$$SST = 382.4510 \text{ m}^2$$

4. ENGINEERING KINEMATICS & INTEGRATION REMARKS

- Peak Inflow Rate:** Evaluated at 0.9600 m/hr , occurring precisely at the inflection hour $t = 29.25 \text{ hrs}$.
- Stage-Time Area Exposure:** Computed as $441.82 \text{ m}\cdot\text{hr}$ via adaptive quadrature compared to $441.80 \text{ m}\cdot\text{hr}$ using the trapezoid rule, representing cumulative hydraulic head exposure across the temporal domain.
- Residual Analysis:** Maximum error magnitude remains tightly bounded at 0.082 m , aligning with physical sensor discretization thresholds without displaying structural model drift or bias.